Chapter 1 – Introduction

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1.1 Introduction

The Intelligent Task Allocation System (ITAS) is a web-based solution designed to optimize software development tasks assigned within IT projects. Traditional manual assignment methods rely heavily on human judgment, which often introduces bias, inefficiency, and inconsistent resource utilization. By leveraging automated analysis of employee CVs, portfolios, and task requirements, ITAS ensures that each task is matched to the most suitable candidate. The motivation behind this project stems from the real challenges faced by software houses, where project managers struggle to identify the right developer for the right task. ITAS addresses this issue by providing a data-driven, fair, and efficient task allocation mechanism.

1.2 Background

In many software houses, Project Managers (PMs) face significant challenges in assigning the right developer to the right task. The fundamental issue lies in the fact that PMs often do not know the full skill set of every employee under their supervision. Consequently, tasks are frequently assigned based on "whoever is available" rather than "who is the best fit," leading to suboptimal resource utilization. Furthermore, junior developers are often overlooked because they lack a proven track record, even if their academic background suggests they possess the specific skills required for a task. This project seeks to bridge this gap by leveraging automated parsing and algorithmic matching.

1.3 Problem Statement

The core challenge addressing the manual task assignment within software development teams can be broken down into the following key areas:

The Knowledge Gap: Project Managers often operate without full visibility into the comprehensive skill sets and emerging capabilities of their team members. Because skills and certifications are rarely centralized or updated in real-time, management often lacks the data necessary to identify which employees possess the specific niche expertise required for complex tasks, leading to blind spots in resource planning.

Inefficient Allocation: Current task assignment processes are frequently driven by immediate availability rather than technical suitability. When tasks are assigned simply because a developer is "free" rather than "fit" for the job, it often results in a mismatch where employees struggle with tasks outside their proficiency. This inefficiency can lead to reduced code quality, slower delivery times, and the need for significant rework.

Underutilization of Talent: There is a systemic tendency to overlook fresh graduates and junior developers due to their lack of professional work history. Despite having relevant academic skills, certifications, or personal portfolios that prove their competence, these individuals are often sidelined in favor of senior staff. This results in a waste of cost-effective talent and prevents junior team members from gaining the experience needed to grow.

Manual Overhead: The process of manually matching specific technical requirements to human resources is both time-consuming and highly subjective. Project managers must

spend excessive time analyzing individual backgrounds or relying on memory to make decisions. This manual overhead introduces human bias and error, diverting valuable time away from strategic leadership and project monitoring.

1.4 Limitations

Time Constraints: The development and refinement of the Intelligent Task Allocation System (ITAS) are strictly bound by the rigid schedule of a graduation project. This fixed timeline limits the opportunity for extensive iterative cycles, prolonged beta testing, or the inclusion of highly complex secondary features, necessitating a focus on core functionality and stability within the academic semester.

Data Availability for Algorithm Training: A significant challenge in developing an accurate matching engine is the requirement for diverse training data. The project has limited access to large-scale datasets of real-world CVs and portfolios due to privacy concerns and academic scope. This data scarcity may affect the algorithm's ability to train on a wide variety of career profiles, potentially impacting the precision of its suitability scoring.

Parsing Complex and Unstructured Formats: Extracting precise skill data is complicated by the lack of standardization in user documents. CVs and portfolios frequently vary in layout, file format, and design complexity. The system faces inherent difficulties in parsing poorly formatted text or creative resume designs, which can lead to challenges in accurately interpreting and indexing an applicant's full skill set.

External API Integration Constraints: To analyze portfolios deeply, the system may need to interface with external platforms such as GitHub. This introduces technical constraints related to API usage, including strict rate limits, authentication barriers, and dependency on third-party data structures. These factors can restrict the speed and depth of the portfolio parsing process.

Exclusion of Soft Skills Evaluation: The ITAS algorithms are designed specifically to evaluate quantifiable technical skills and hard qualifications. Consequently, the system cannot assess subjective but critical human attributes such as communication style, leadership potential, or teamwork. This remains a gap that requires human intervention, as the software focuses solely on technical compatibility rather than cultural fit.

Soft Skills: The system focuses on technical skills and cannot fully evaluate soft skills such as

communication or teamwork.

1.5 Project Objectives

Development of a Core Web-Based Automation System To design and develop a secure, web-based application that automates the manual task assignment process. The objective is to deliver a functional, robust prototype within the graduation project timeline, providing a

centralized platform that replaces disjointed manual workflows with a streamlined digital environment.

NLP-Powered Technical Skill Extraction To build an NLP-powered engine capable of parsing and extracting data from employee CVs and portfolios. Recognizing the challenge of varied document formats, the system aims to standardize unstructured text into a structured skill repository, specifically focusing on identifying hard technical skills and qualifications to overcome the exclusion of soft skill analysis.

Algorithmic Suitability Scoring To implement a matching algorithm that calculates suitability scores for employees by comparing their verified technical skills against specific task requirements. This data-driven approach aims to bridge the "knowledge gap" by objectively ranking candidates, ensuring that assignments are based on relevant technical merit rather than just availability.

Workload and Availability Visualization To provide a comprehensive dashboard that visualizes workload distribution and real-time employee availability. This tool is designed to support Project Managers in making informed decisions, helping to prevent burnout and underutilization by presenting clear, actionable data on team capacity.

Enhancing Fairness through Data-Driven Allocation To enhance fairness and efficiency in task allocation by minimizing human bias and subjectivity. By relying on automated skill matching rather than personal preference or memory, the system seeks to provide equal opportunities for all team members, including fresh graduates who possess the necessary academic skills but lack extensive professional history.

1.6 Project Solution Overview

The proposed solution is a web application that automates software task assignments using a multi-step intelligent process. The system parses CVs and portfolio data, identifies relevant skills using NLP and pattern matching, and compares them with the skill tags defined for each task. A scoring algorithm ranks employees based on suitability, enabling project managers to make data-driven decisions. The dashboard provides real-time visibility into task progress and employee workload, ensuring balanced and efficient project flow.

1.7 Project Scope

• **Scope Covered:**

– CV and portfolio text extraction.

– NLP-based skill identification and classification.

– Task creation with required skill tags.

– Algorithmic matching and suitability scoring.

– Dashboard for project managers.

– Employee profile management.

• **Scope Not Covered:**

– Automated evaluation of soft skills.

– Full integration with third-party HR systems.

– Real-time monitoring of coding performance.

– Mobile application development.

1.8 Project Feasibility

• **Technical Feasibility**: The system uses widely available and well-supported technologies such as Python, Django/Flask, React.js, PostgreSQL, PyPDF2, and spaCy. All required tools are accessible and align with the team's skill set.

• **Operational Feasibility**: ITAS enhances the workflow of software project managers and supports organizational goals by improving fairness, efficiency, and transparency in task allocation.

• **Economic Feasibility**: Development costs are minimal as the system relies on open-source tools and student resources. No commercial licensing is required, making ITAS cost-efficient and practical for academic and small organizational use.